

The ACM Multimedia 2025 Grand Challenge of Avatar-based Multimodal Empathetic Conversation

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Abstract

The Ava-MERG Challenge at ACM Multimedia 2025 explores avatar-based multimodal empathetic response generation by advancing dialogue systems that respond with emotional awareness across text, speech, and talking-face video. To support this goal, we constructed a high-quality dataset featuring aligned multimodal responses in diverse conversational scenarios. The challenge comprises two progressively complex subtasks centering around multimodal-aware empathetic response generation. We aim to promote research in empathetic AI and affective computing. More information is available at <https://AvaMERG.github.io/MM25-challenge>.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Empathetic Response Generation, Affective Computing

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1 Introduction

While Large Language Models (LLMs) [2–4, 6, 11, 14, 19, 20] advance machine intelligence, true AGI requires emotional and empathetic abilities. To overcome the limits of text-only Empathetic Response Generation (ERG) [13], we propose the Avatar-based Multimodal Empathetic Response Generation (Ava-MERG) benchmark [5, 21], which requires synchronized responses across text, speech, and facial video. Its accompanying challenge fosters stronger multimodal

Table 1: Statistics of AvaMERG dataset.

	Item	Stats
Dialogue	#Train Set	24,696
	#Valid Set	4,373
	#Test Set	3,979
	#Total	33,048
	Avg. Words Per Utterance	14.68
Modality	Avg. Utterance Per Dialogue	4.6
	Utterance Text	152,021
	Speech Audio	152,021
	Talking-head Video	152,021
	Avg. Length (Sec) Per Aud/Vid	5.67
Avatar	Child (Male/Female)	3/3
	Young (Male/Female)	25/17
	Middle-aged (Male/Female)	4/4
	Elderly (Male/Female)	5/4
	Tone (Emphatic/Mild/Gentle)	14/38/13
Emotion	Race	5
	Text/Multimodal	32/7
Topic&Scenario		10

emotion modeling with applications in assistants, healthcare, and education.

2 Dataset Specification

We build the AvaMERG dataset by enriching the Empathetic Dialogue (ED) corpus [13] with multimodal and identity annotations. Speaker profiles (age, gender, tone, topic) are generated via GPT-4, and additional dialogues are created to balance demographics. Volunteers then record speech and talking-head videos aligned with these dialogues and profiles.

3 Challenge Overview and Evaluation

This challenge aims to build systems that understand multimodal dialogue contexts and generate emotionally appropriate responses across multimodalities [8–10, 15–18], e.g., text, speech, and video. It consists of two subtasks:

Subtask I: Multimodal-Aware Empathetic Response Generation. Participants develop models that generate empathetic text and speech responses from multimodal dialogue contexts. Evaluation combines automatic metrics—emotion prediction accuracy (**Acc**) and diversity (**Dist-1/2**)—with human assessments of empathy and fluency (**HE**).



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Subtask II: Multimodal Empathetic Response Generation.

Here, models must additionally produce talking-face videos aligned with generated text and speech. Besides metrics from Subtask I, human judges also evaluate Multimodal Coherence (MC), Naturalness (NT), and Emotional Expressiveness (EE).

Table 2: Task1 Top-performing Results

Rank	Team	Acc. (%)	Dist-1	Dist-2	HE
1	It's MyGO!!!!	68.42	3.93	23.64	3.8
2	DZ	52.75	3.06	20.17	4.2
3	AI4AI	41.79	4.79	31.79	4.1

Table 3: TASK2 Top-performing Results

Rank	Team	MC	NT	EE	Total
1	SYSU_RUNNER	4.0	3.8	4.3	12.1
2	AI4AI	3.8	4.1	3.6	11.5
3	It's MyGO!!!!	3.5	3.2	3.5	10.2

4 Top-performing Methods

Tables 2 and 3 present the top-performing results for Subtask 1 and Subtask 2, respectively. Below, we briefly introduce the approaches adopted by these leading methods.

Team It's my Go!!! employs Vicuna [1] to generate empathetic textual responses guided by structured cues and a chain-of-empathy reasoning process. The generated text is then converted into expressive speech using StyleTTS2 [12] and rendered into synchronized facial video through an Emotionally Aligned Talking-face (EAT) [7] generator.

Team SYSU_RUNNER proposes E3RG, an Explicit Emotion-driven Empathetic Response Generation system built upon multimodal LLMs. E3RG addresses the AvaMERG task by decomposing it into three stages: empathy understanding, memory retrieval, and response generation, further enhanced by expressive speech and video synthesis modules.

TEAM AI4AI presents EMO-Avatar, an emotion-support agent that integrates affective reasoning with multimodal expressiveness. It features a LLM-driven three-stage emotional support strategy and generates empathetic responses across multiple modalities, including voice, gestures, facial expressions, and actions.

TEAM DZ proposes the MERIA framework, which leverages a β -VAE-based multimodal disentanglement encoder to effectively mitigate emotional inconsistencies across different modalities and extend the chain of empathy in the AvaMERG dataset to multimodal aspects.

5 Conclusion

We successfully organized the Avatar-based Multimodal Empathetic Conversation Grand Challenge at ACM MM 2025. The challenge attracted broad participation, and the top three teams for each subtask were selected based on their performance. We hope that the AvaMERG Challenge, along with the released benchmark and tools, will foster continued research and innovation in multimodal empathetic response generation.

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